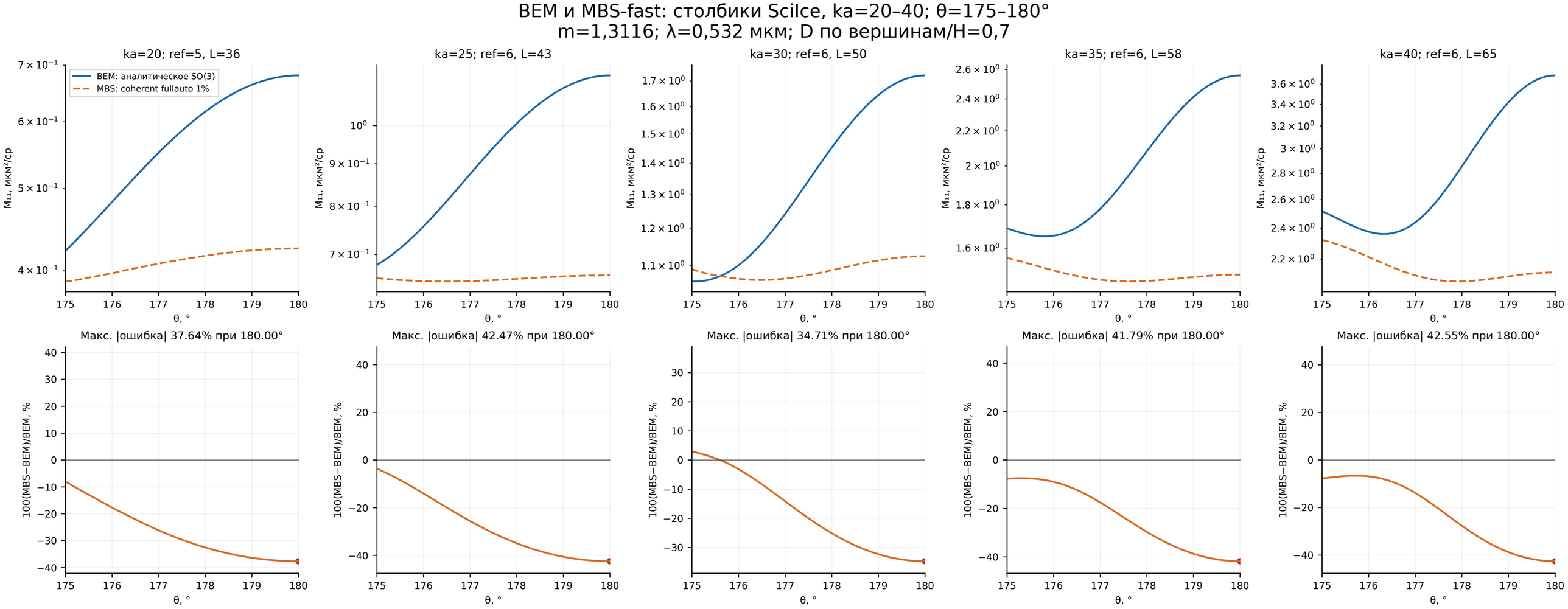
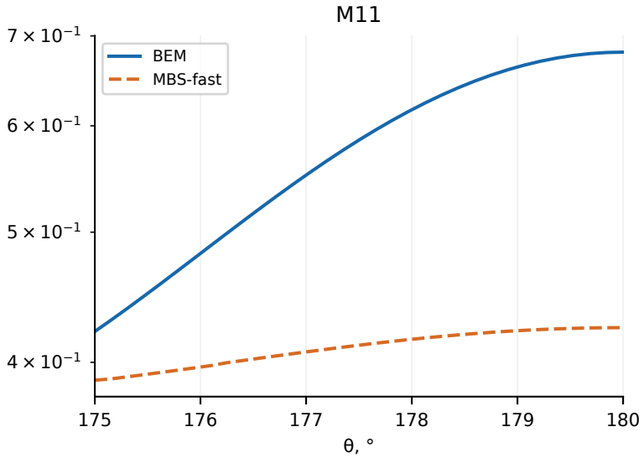
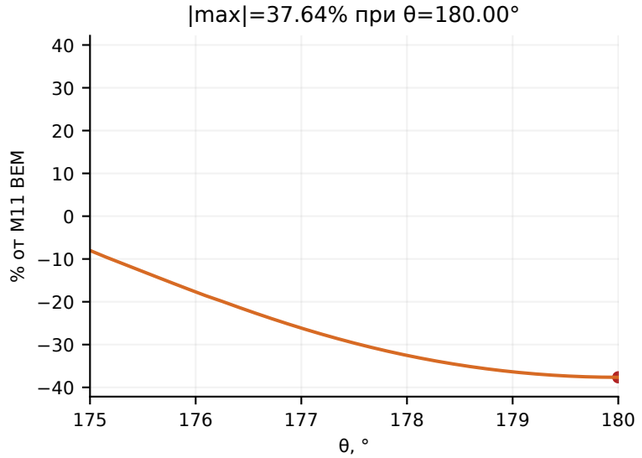
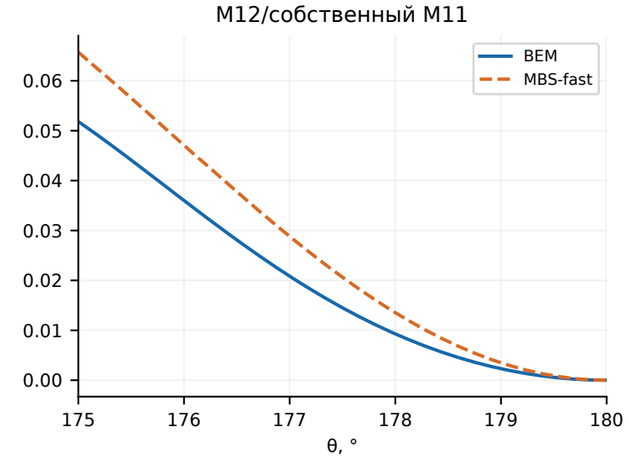

BEM и MBS-fast: столбики Scilce, ka=20–40; θ=175–180°
m=1,3116; λ=0,532 мкм; D по вершинам/H=0,7
ka=20; ref=5, L=36
ka=25; ref=6, L=43
ka=30; ref=6, L=50
ka=35; ref=6, L=58
ka=40; ref=6, L=65
BEM: аналитическое SO(3)
MBS: coherent fullauto 1%
M11, мкм²/ср
θ, °
Макс. |ошибка| 37.64% при 180.00°
Макс. |ошибка| 42.47% при 180.00°
Макс. |ошибка| 34.71% при 180.00°
Макс. |ошибка| 41.79% при 180.00°
Макс. |ошибка| 42.55% при 180.00°
100(MBS−BEM)/BEM, %

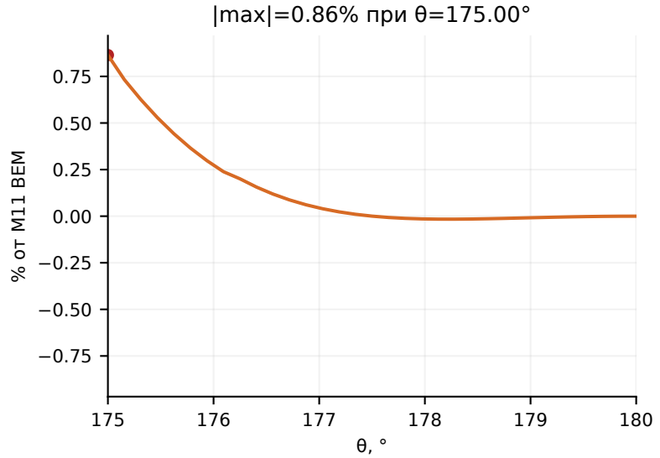
ka=20: все ненулевые элементы и ошибки

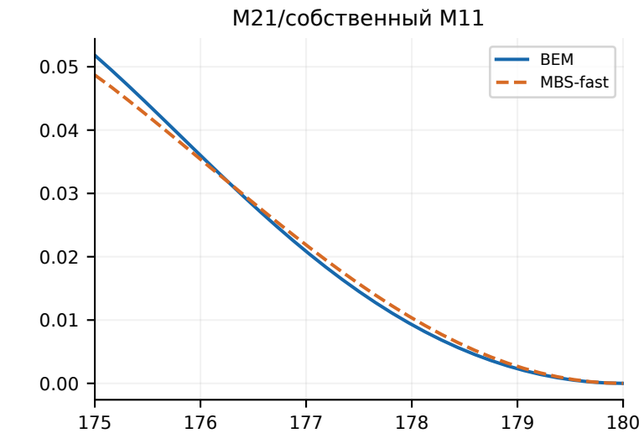
M_{11} размерный; остальные M_{ij}/M_{11} каждого метода. Ошибка: $100(M_{ij}\ \mathrm{MBS} - M_{ij}\ \mathrm{BEM})/M_{11}\ \mathrm{BEM}$

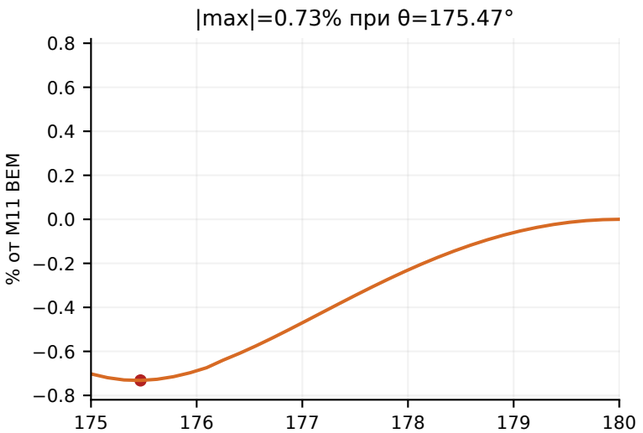


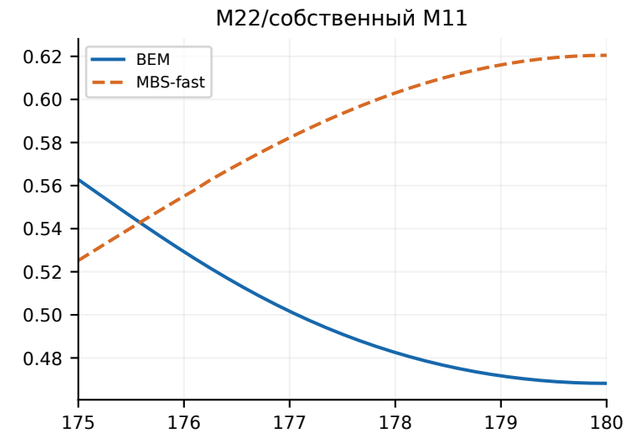


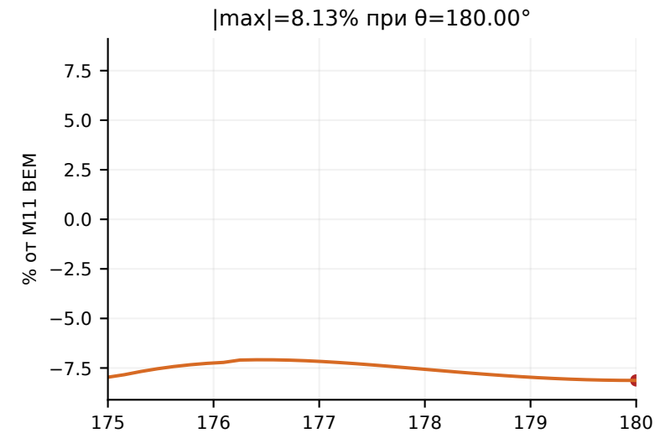


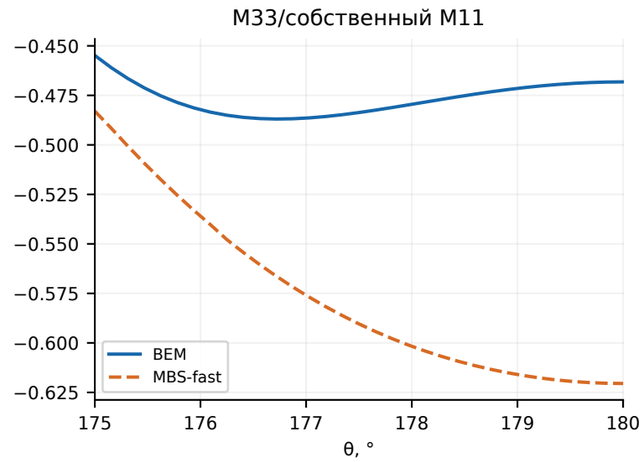


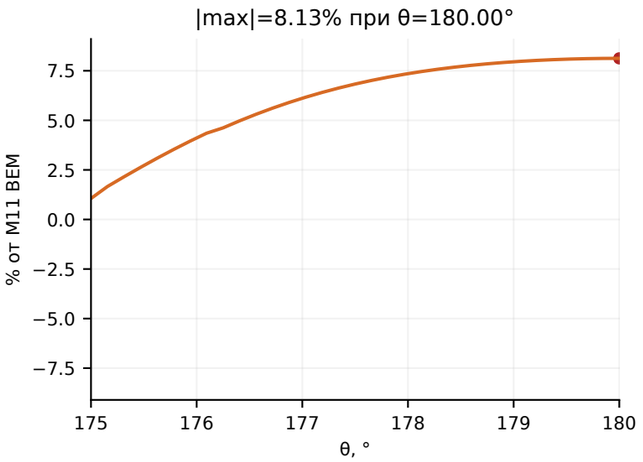


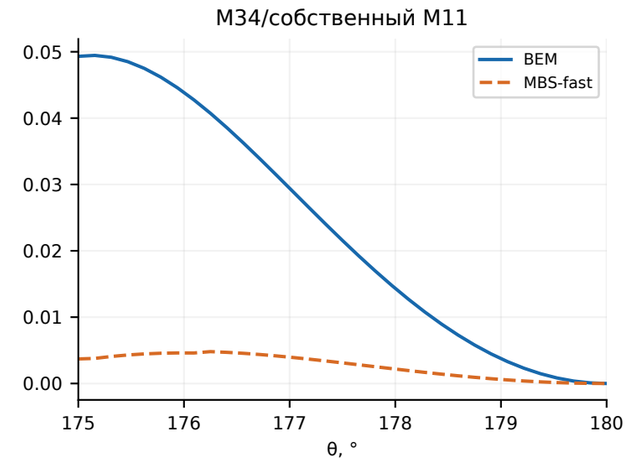


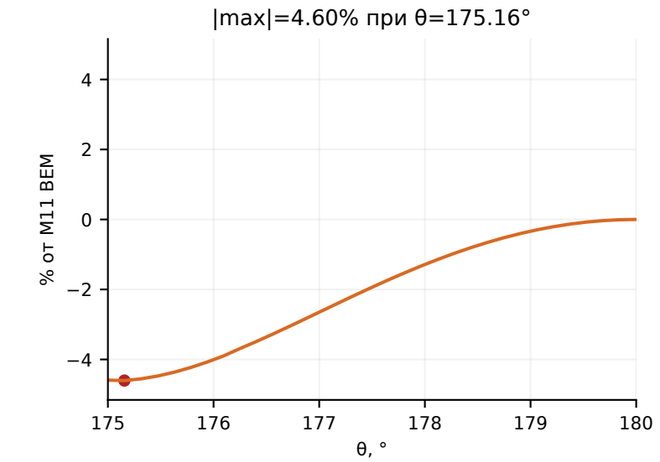


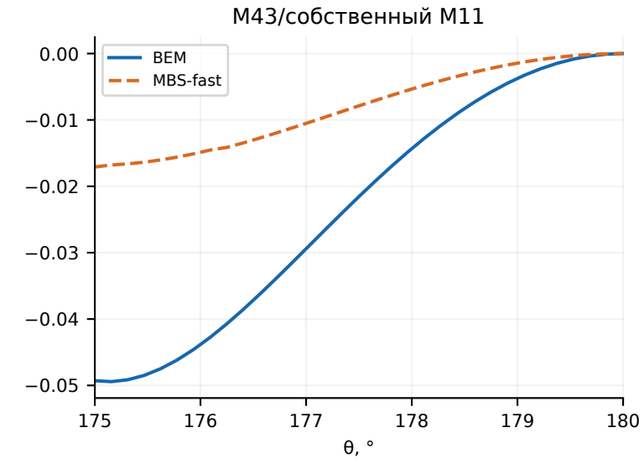


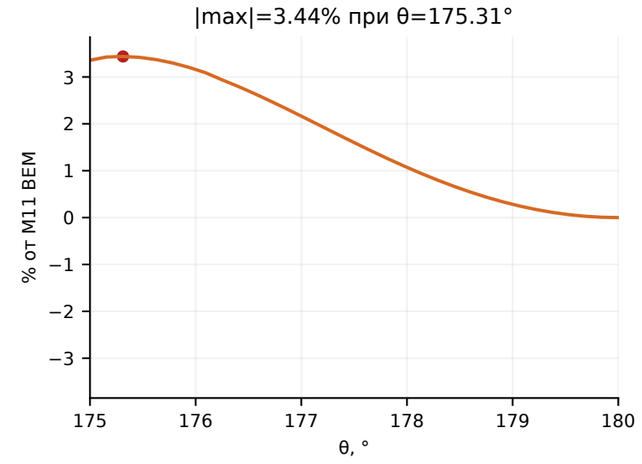


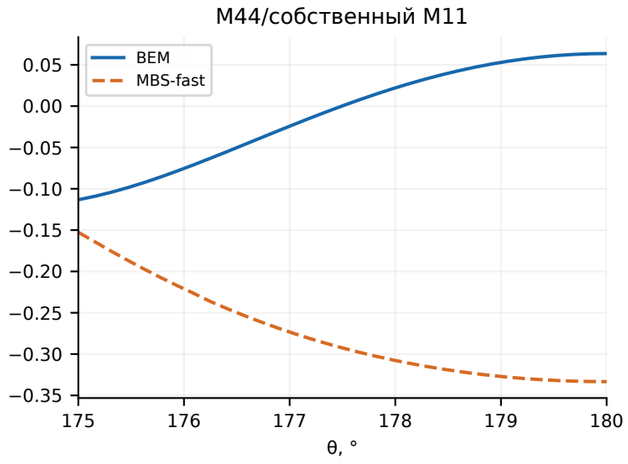


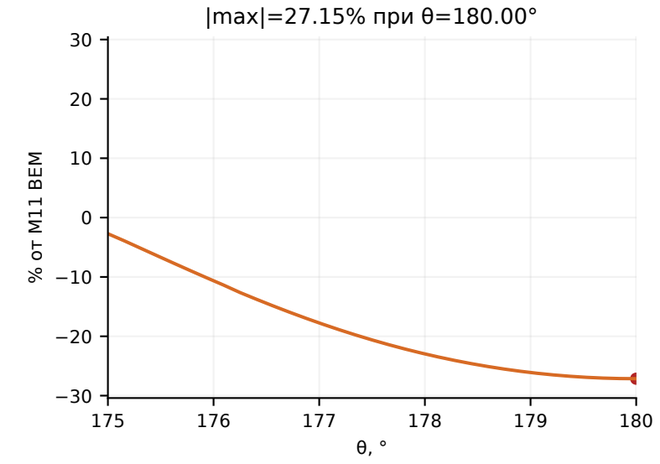






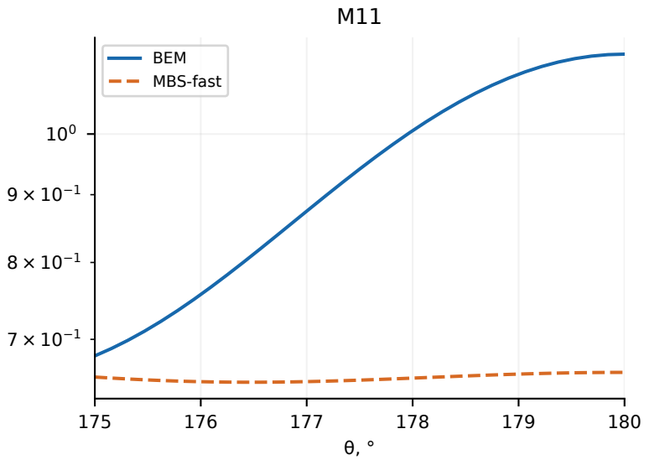


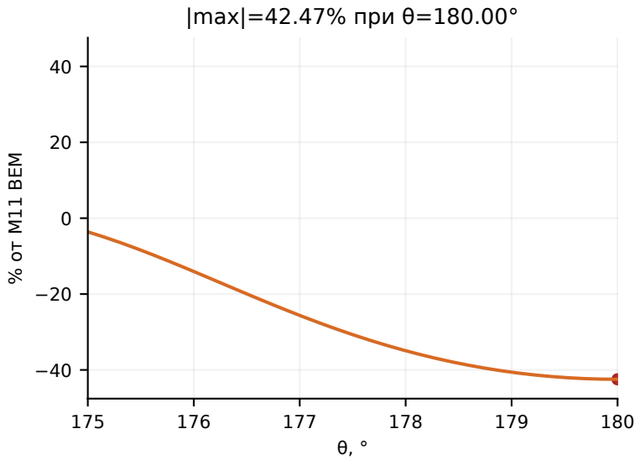


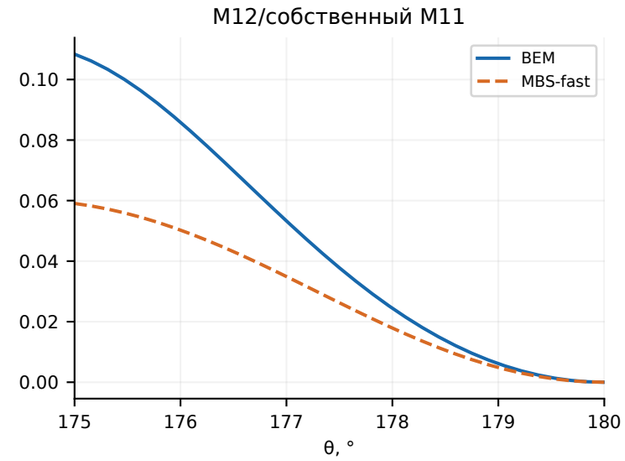


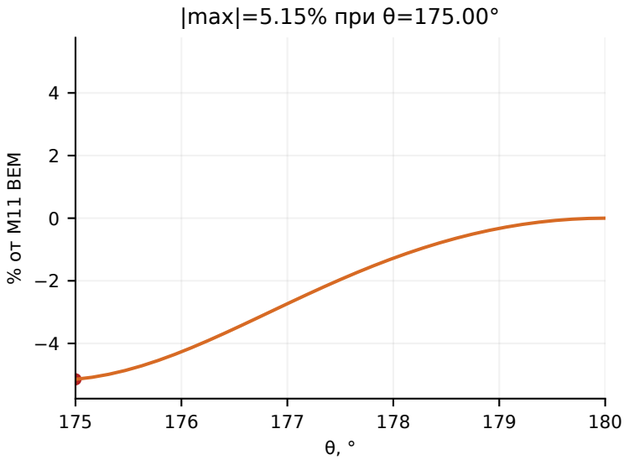
ka=25: все ненулевые элементы и ошибки

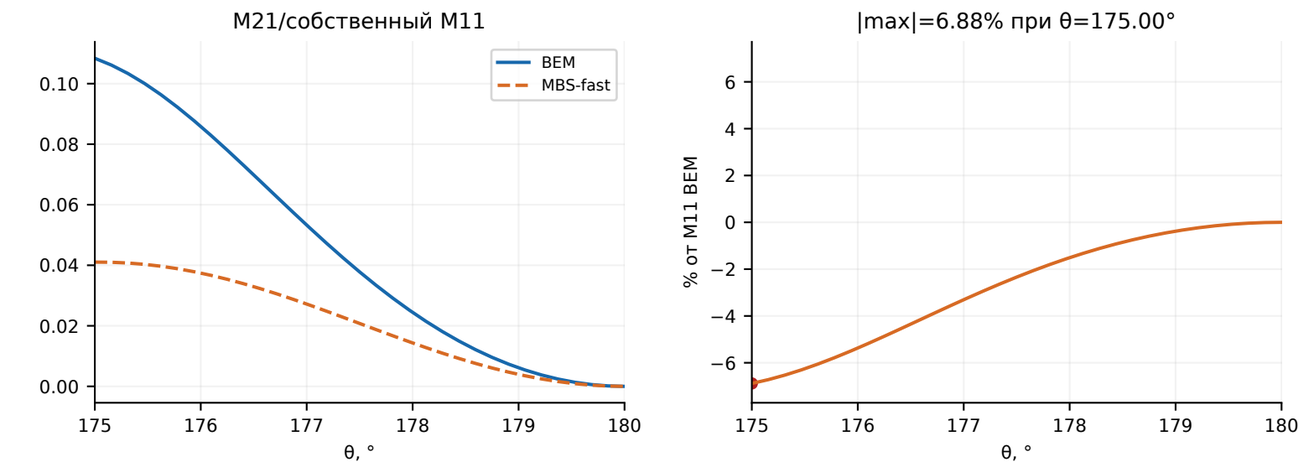
M_{11} размерный; остальные M_{ij}/M_{11} каждого метода. Ошибка: $100(M_{ij}$ MBS − M_{ij} BEM$)/M_{11}$ BEM

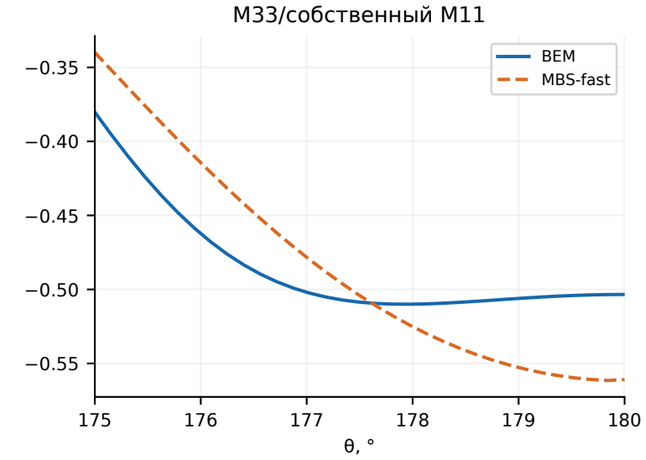


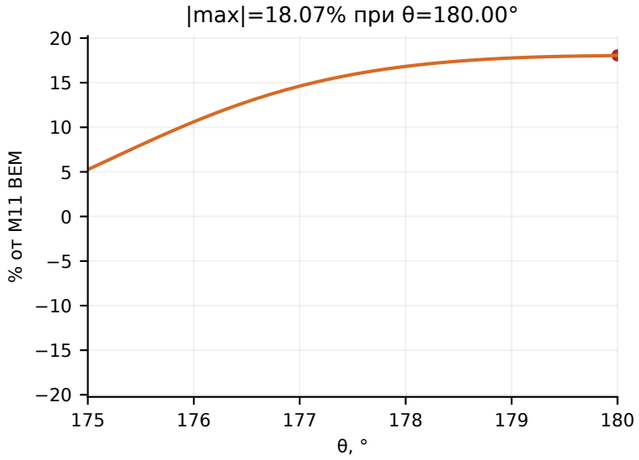


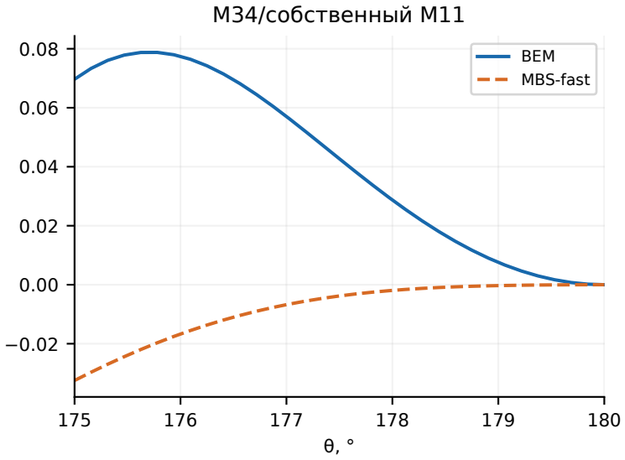


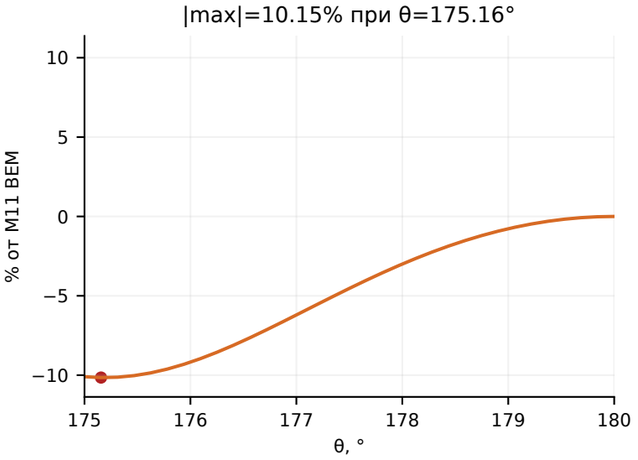


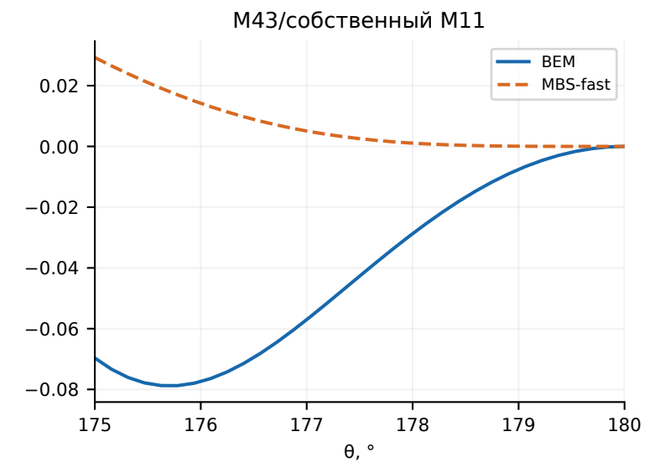


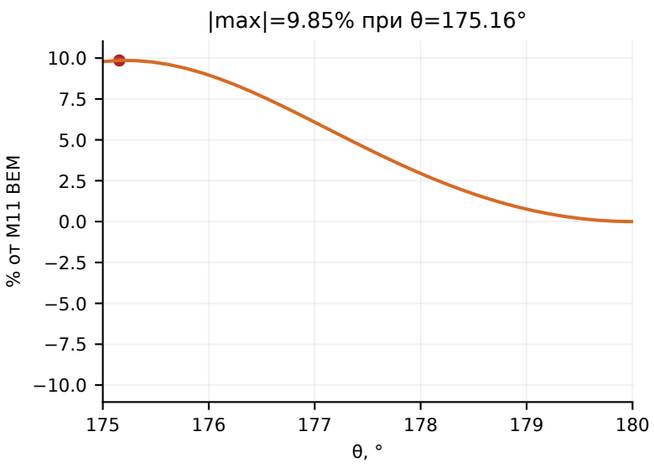


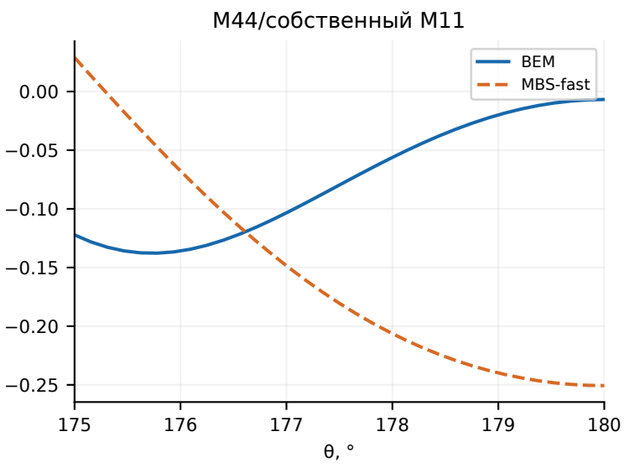


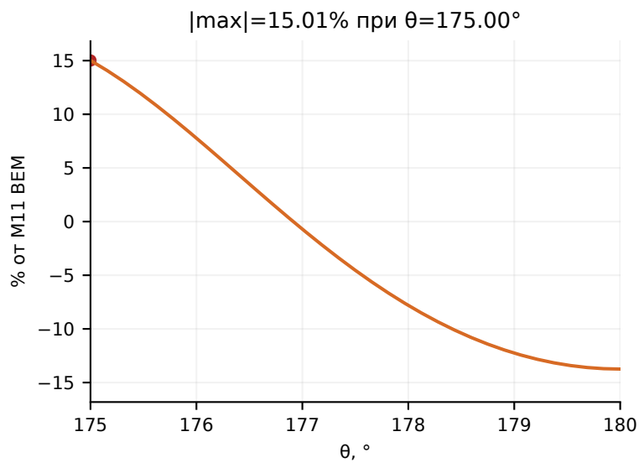






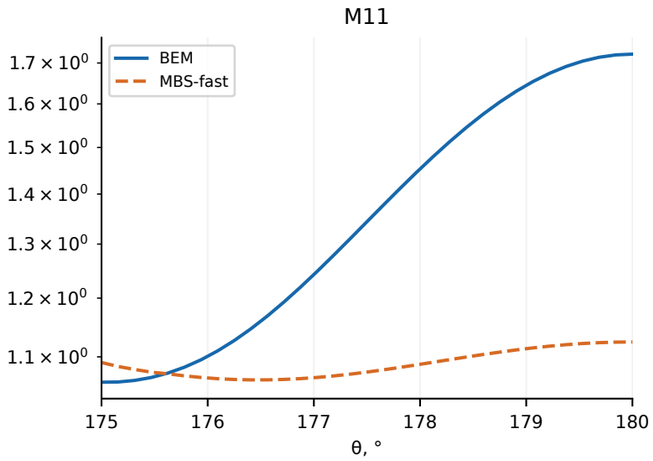


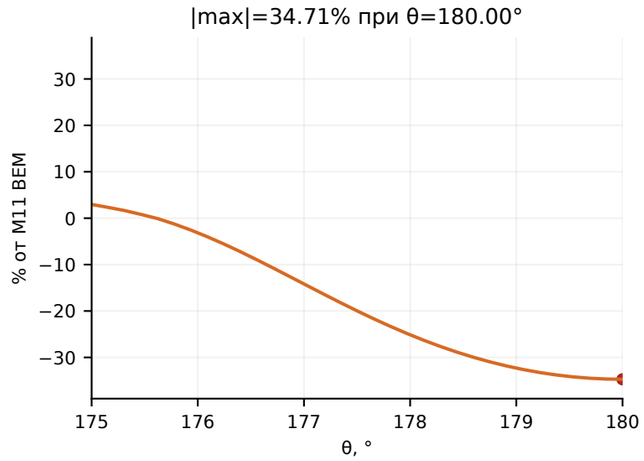


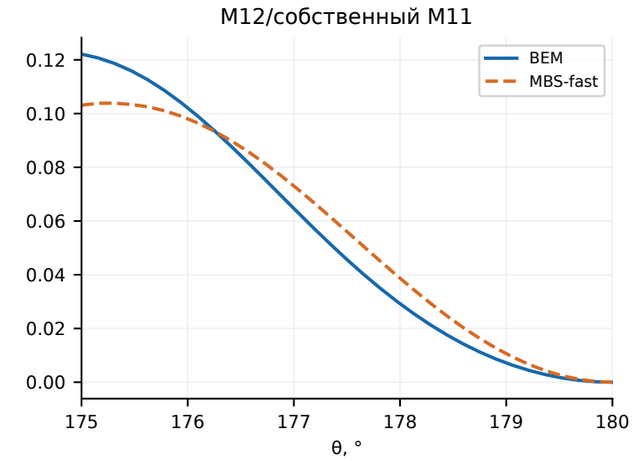


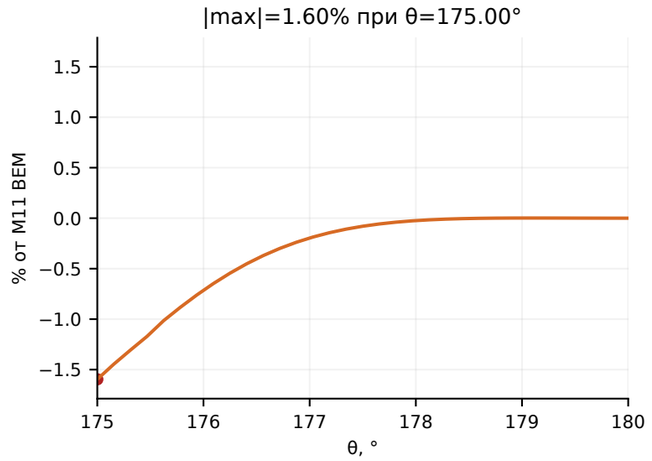
ka=30: все ненулевые элементы и ошибки

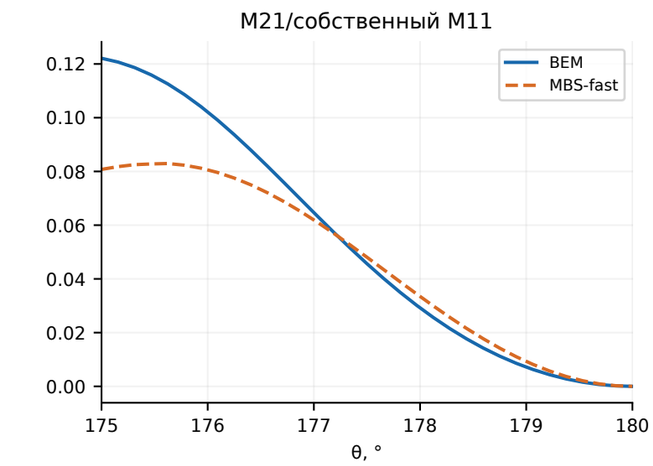
M_{11} размерный; остальные M_{ij}/M_{11} каждого метода. Ошибка: $100(M_{ij}\ \mathrm{MBS} - M_{ij}\ \mathrm{BEM})/M_{11}\ \mathrm{BEM}$

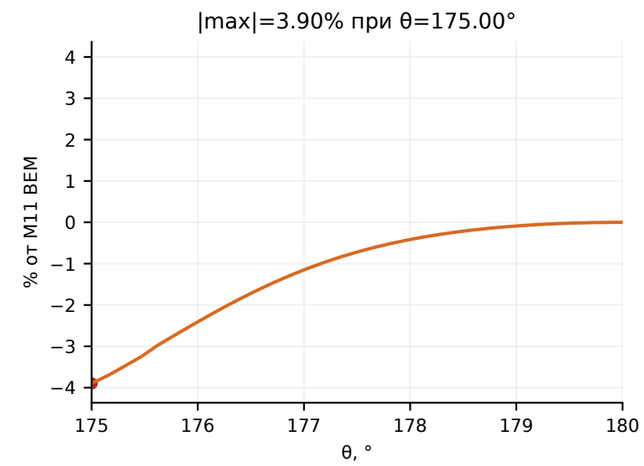


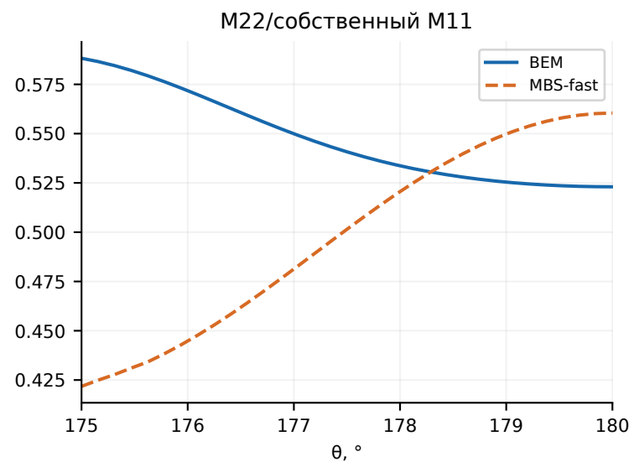


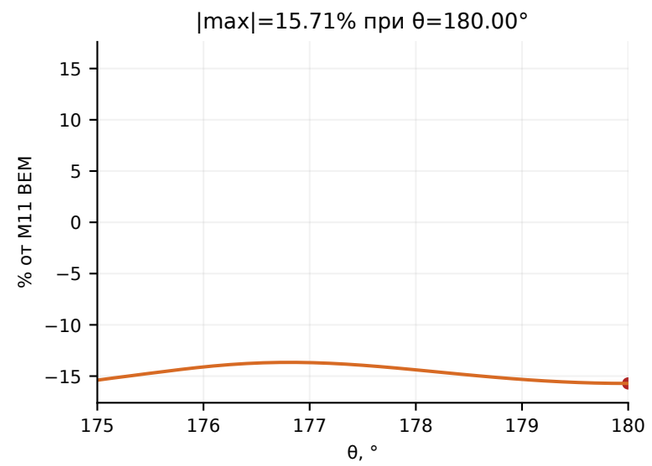


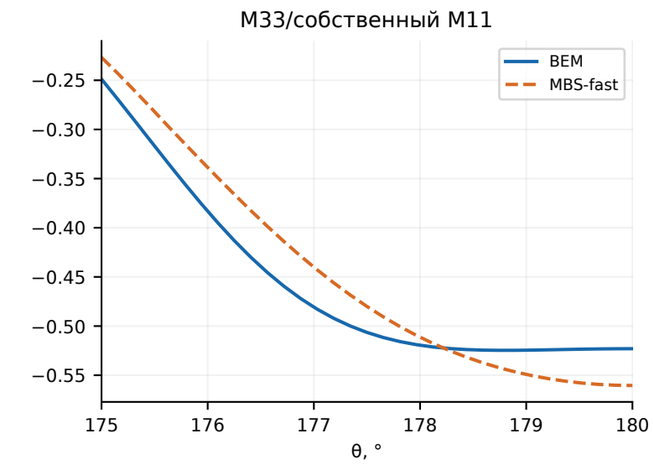


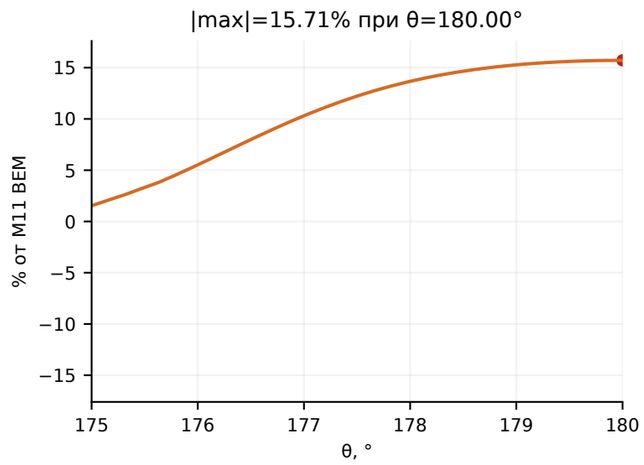


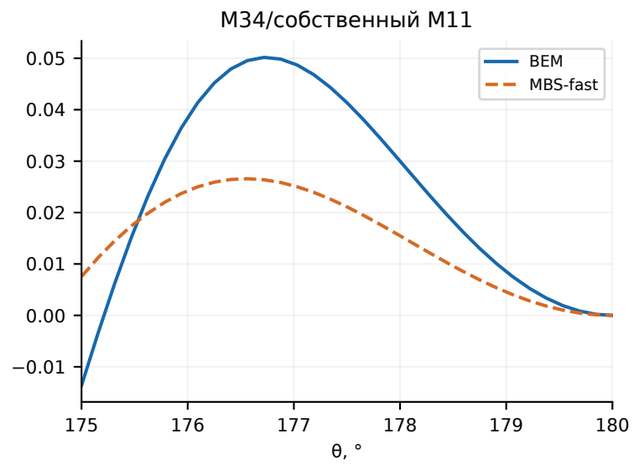


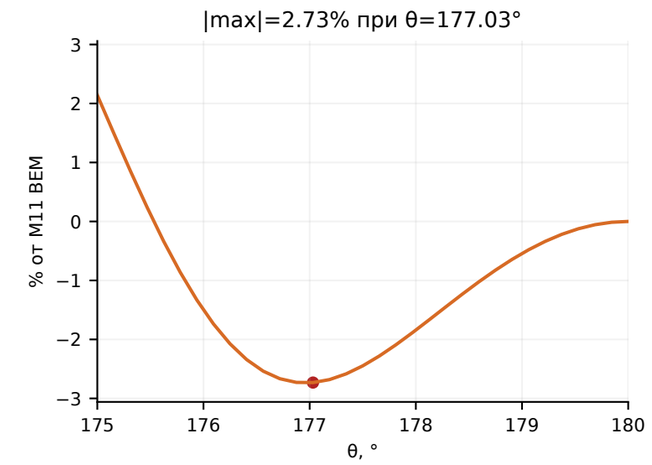


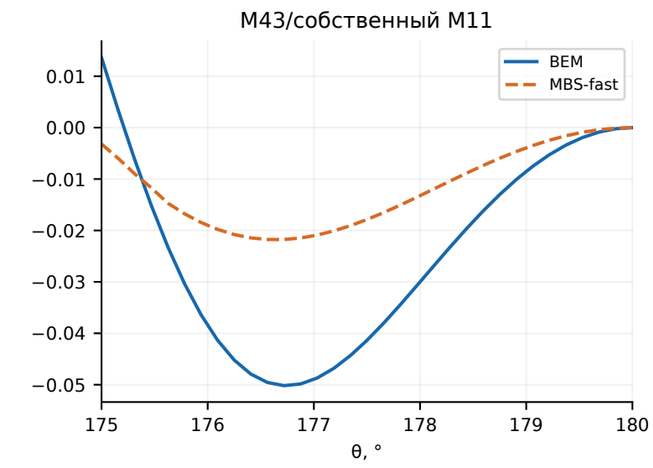


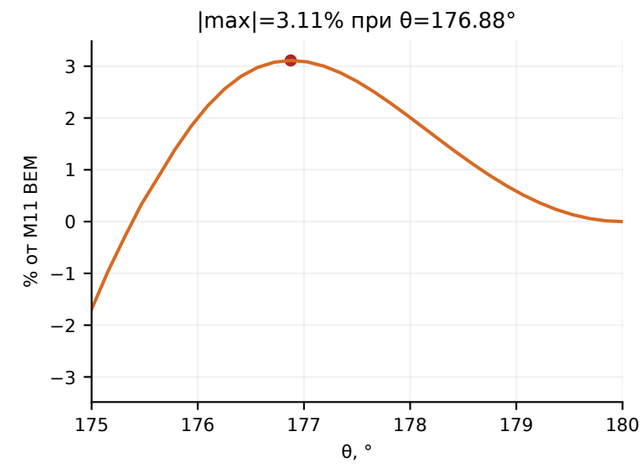


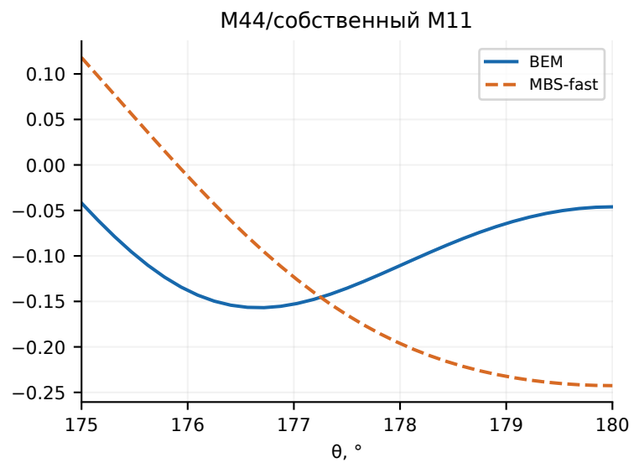


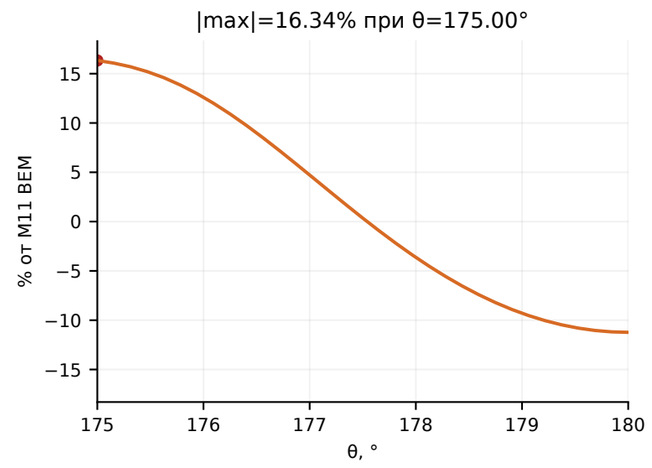






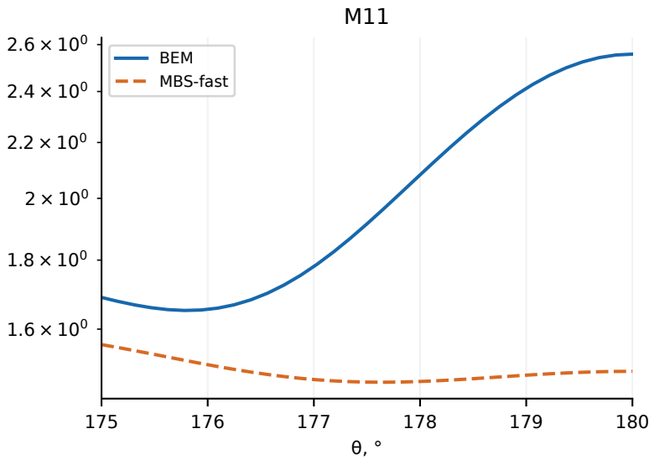


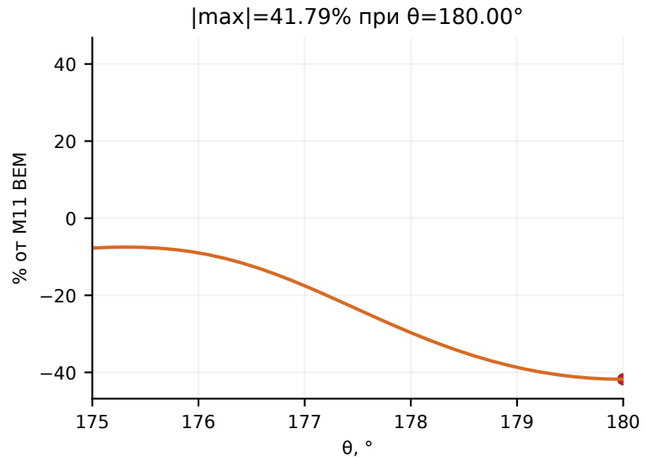


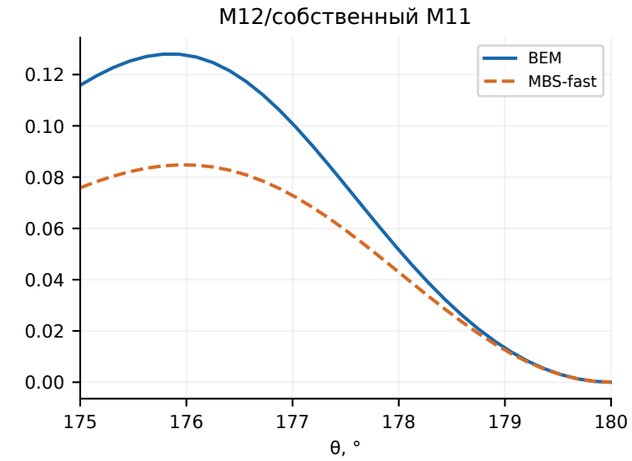


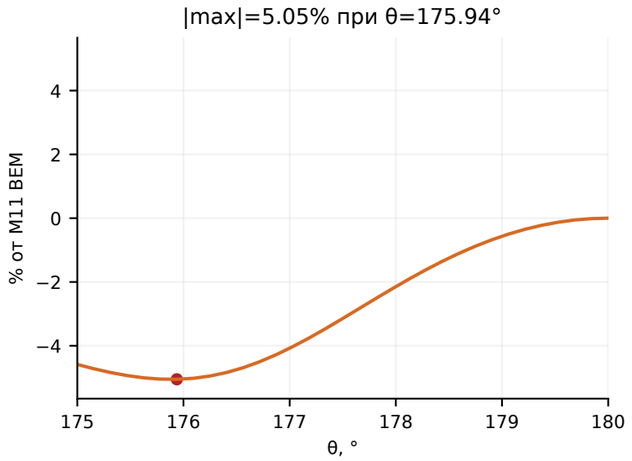
ka=35: все ненулевые элементы и ошибки

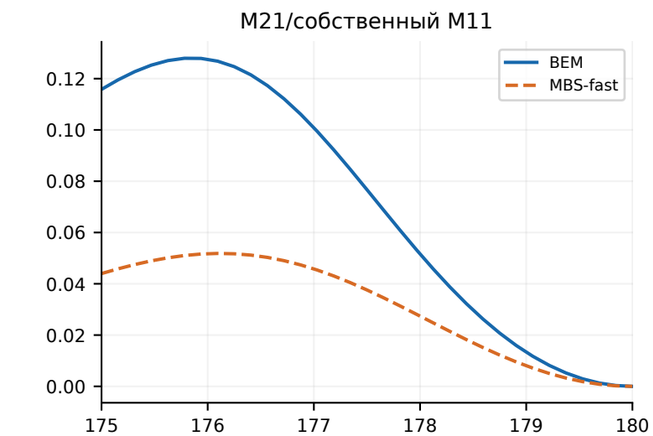
M_{11} размерный; остальные M_{ij}/M_{11} каждого метода. Ошибка: $100(M_{ij}$ MBS − M_{ij} BEM$)/M_{11}$ BEM

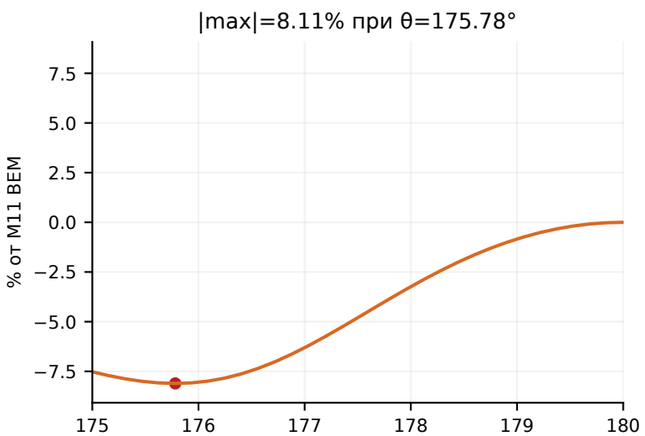


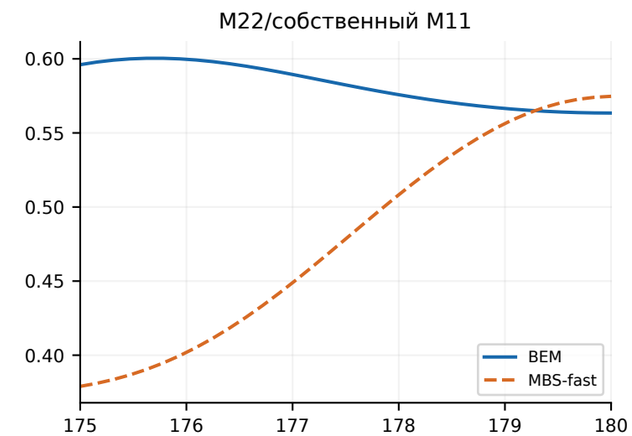


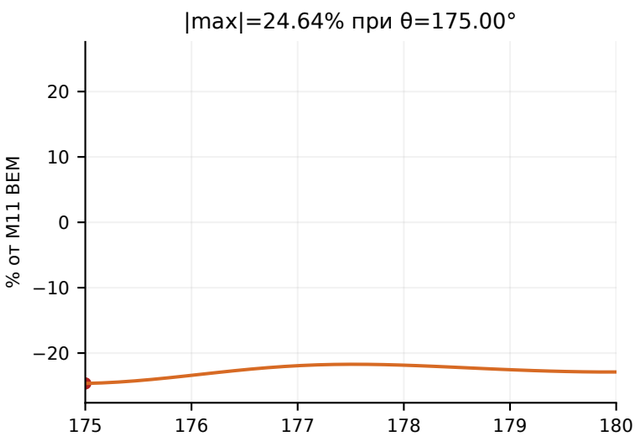


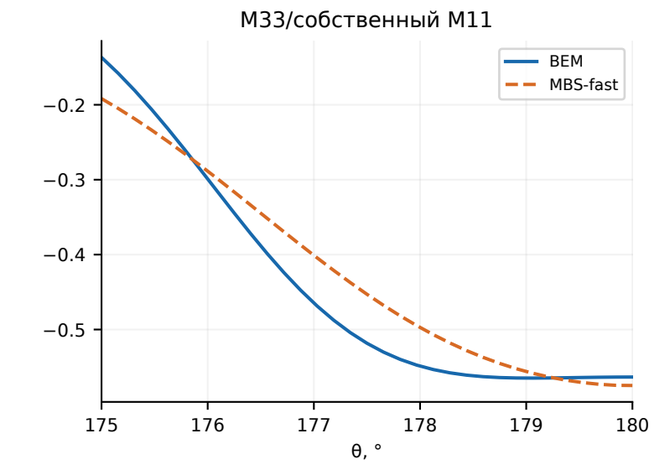


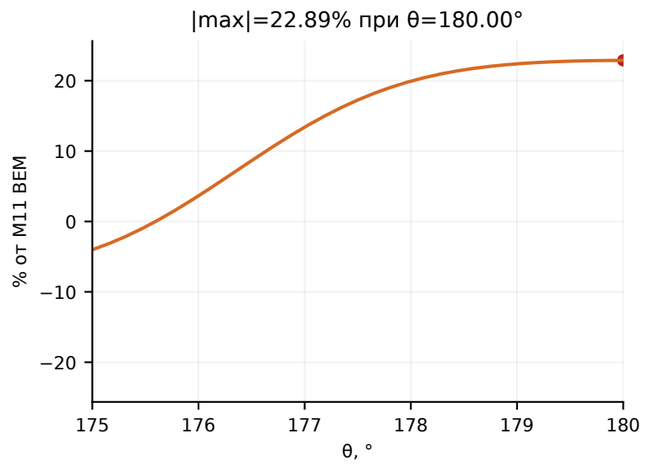


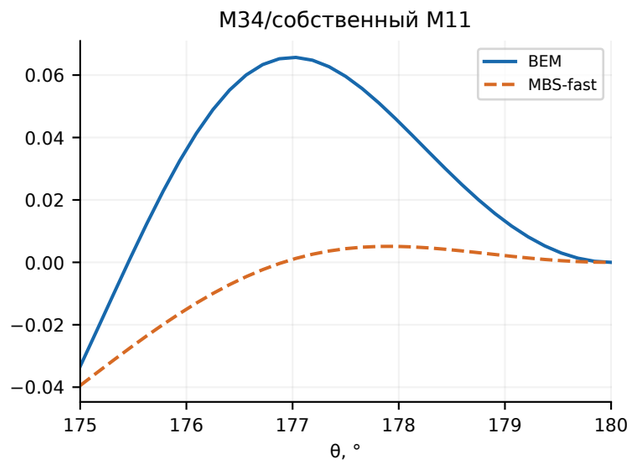


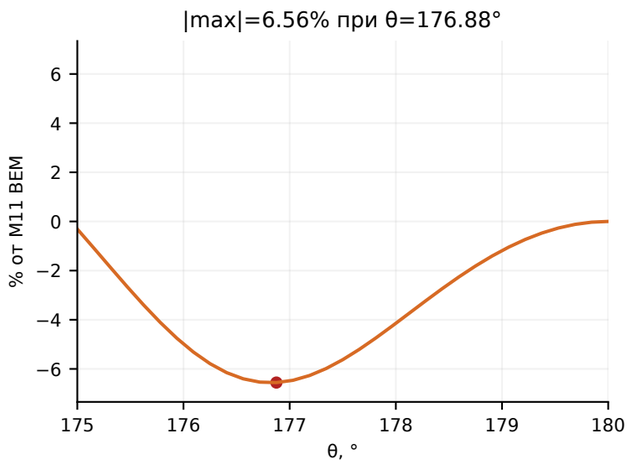


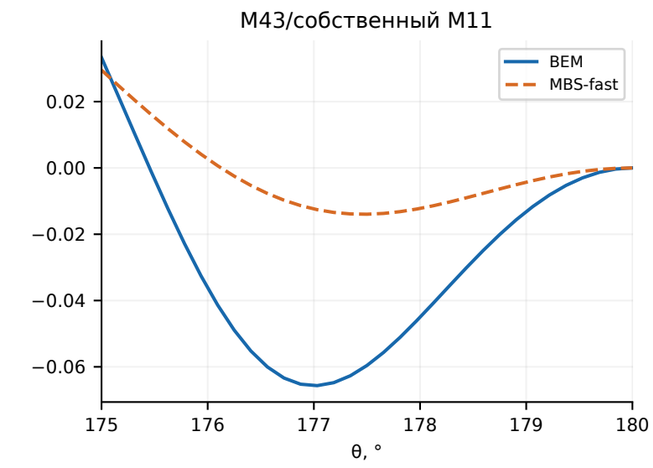


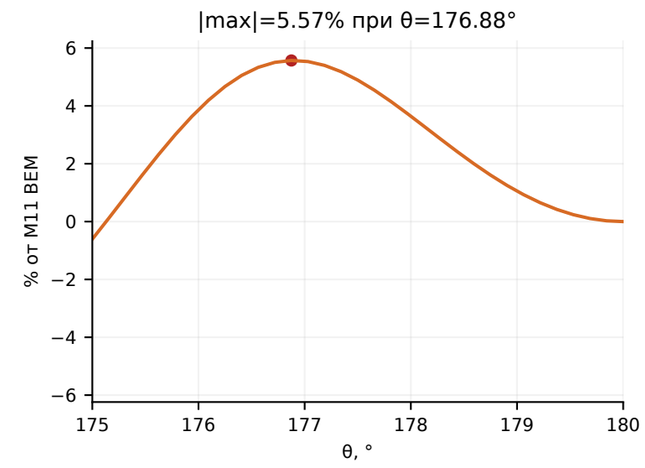


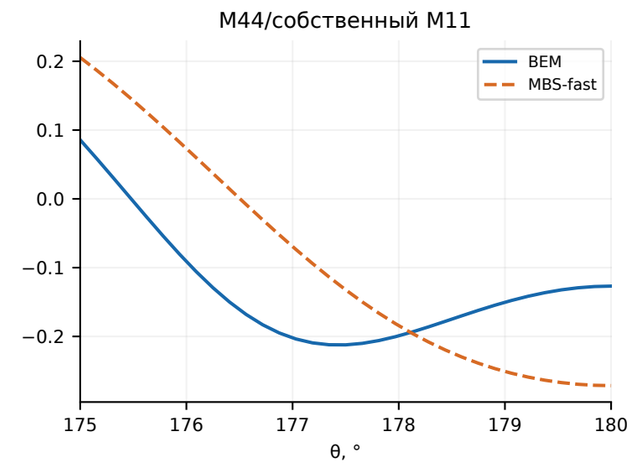


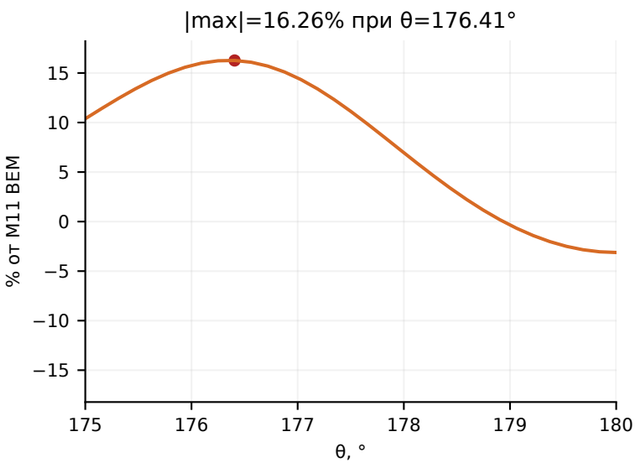






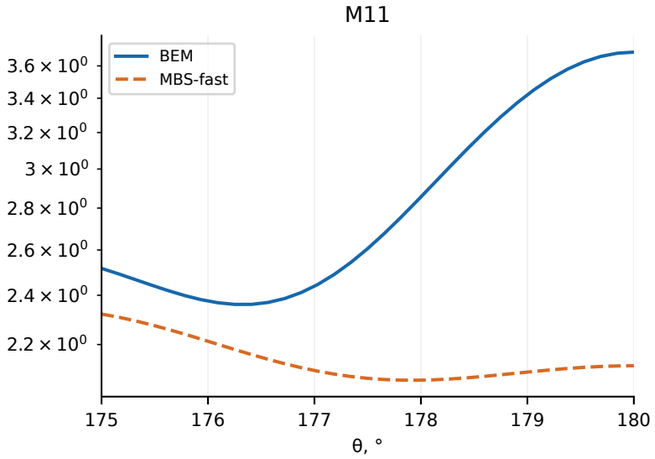


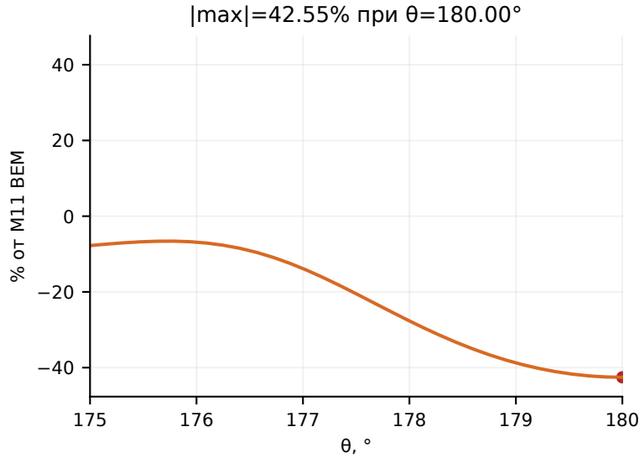


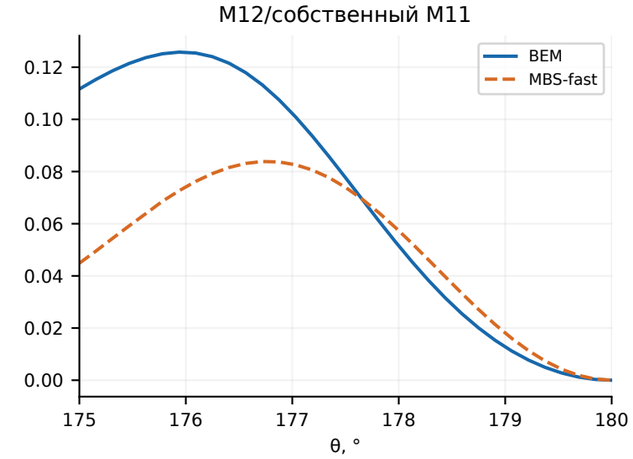


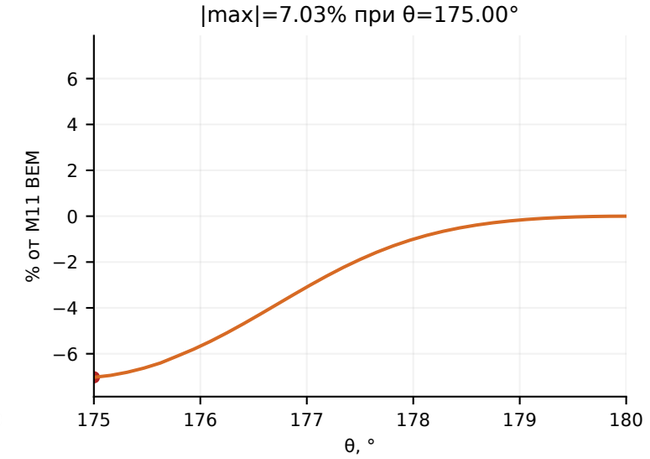
ka=40: все ненулевые элементы и ошибки

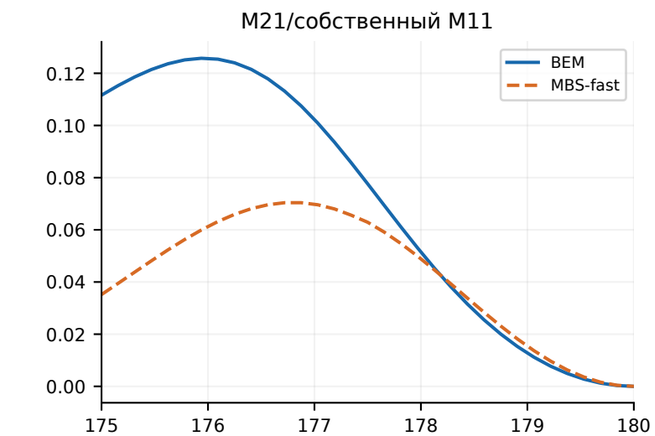
M_{11} размерный; остальные M_{ij}/M_{11} каждого метода. Ошибка: $100(M_{ij}\ \mathrm{MBS} - M_{ij}\ \mathrm{BEM})/M_{11}\ \mathrm{BEM}$

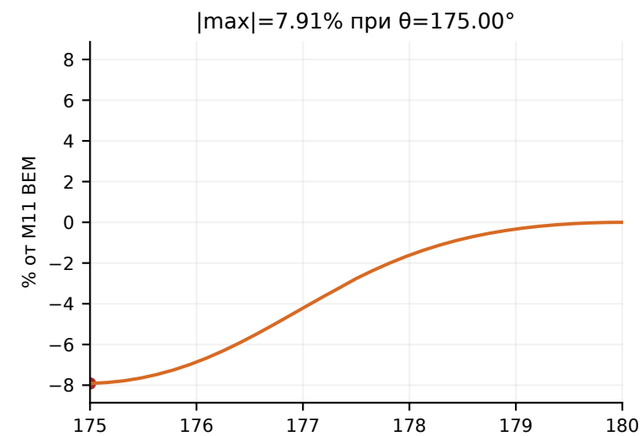


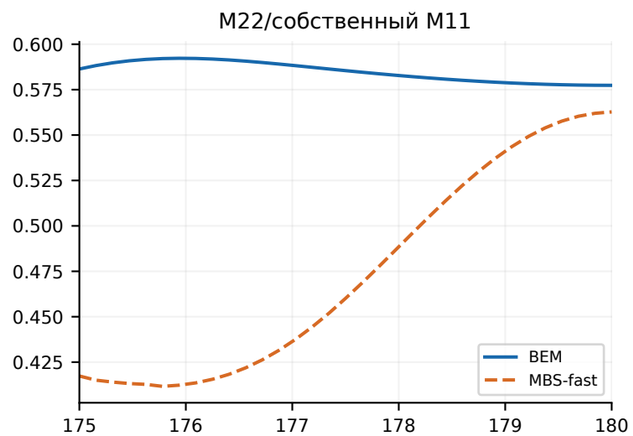


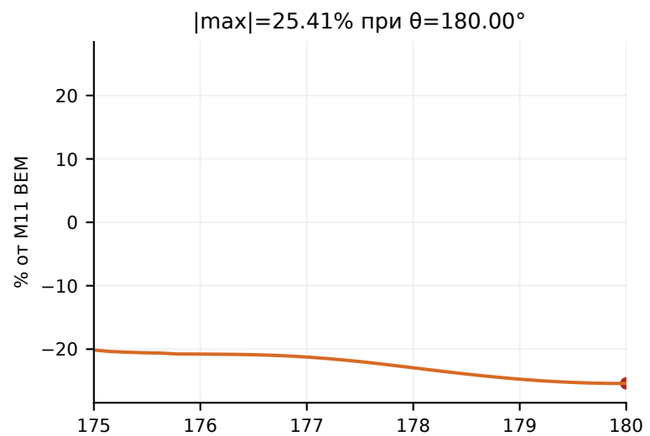


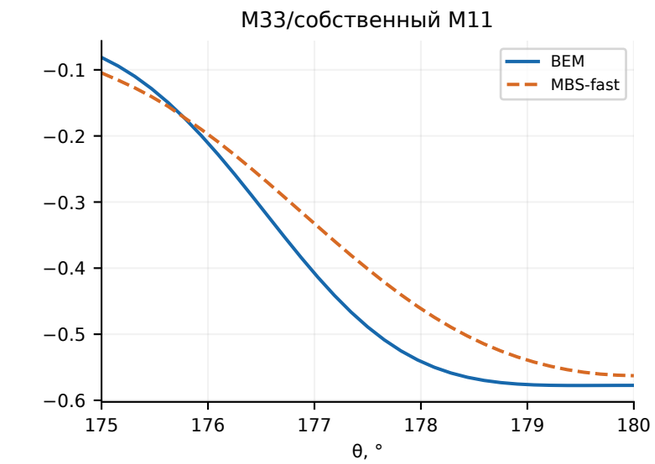


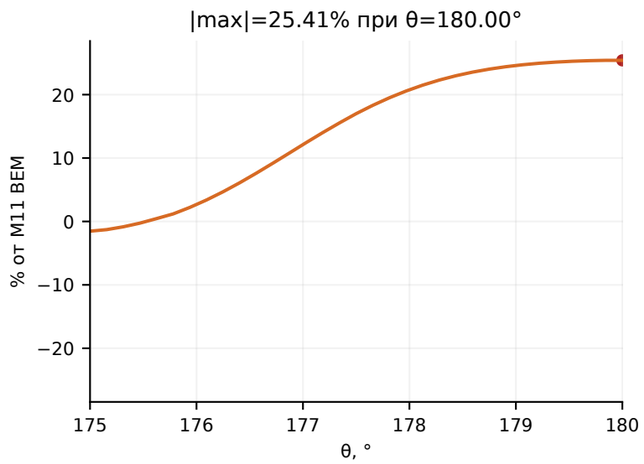


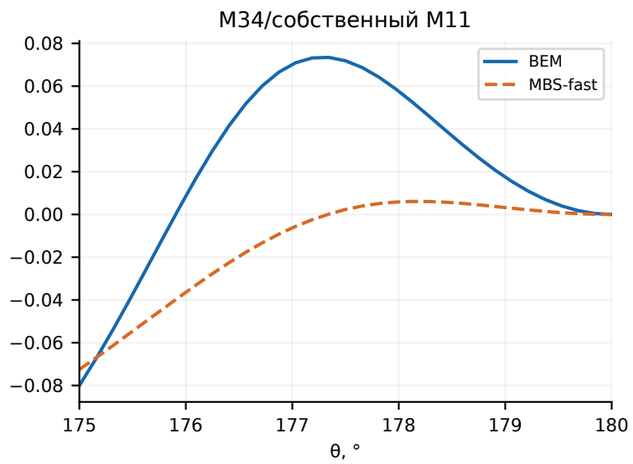


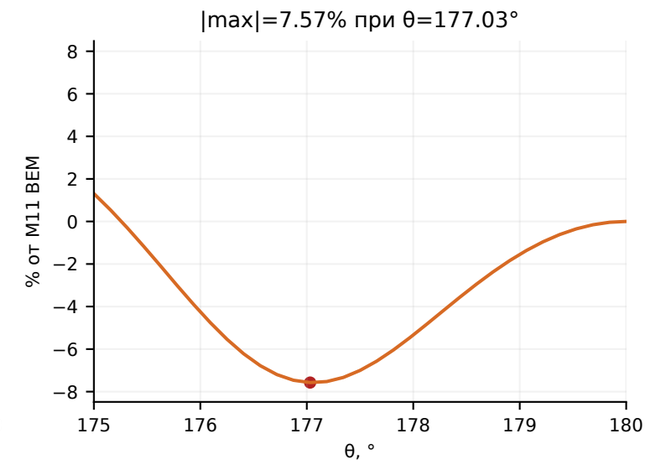


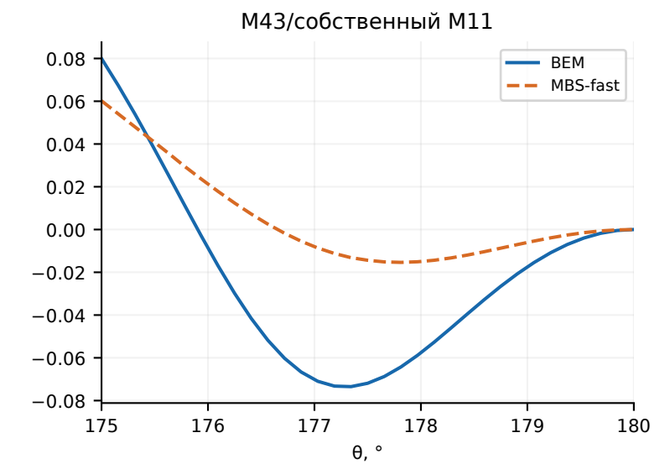


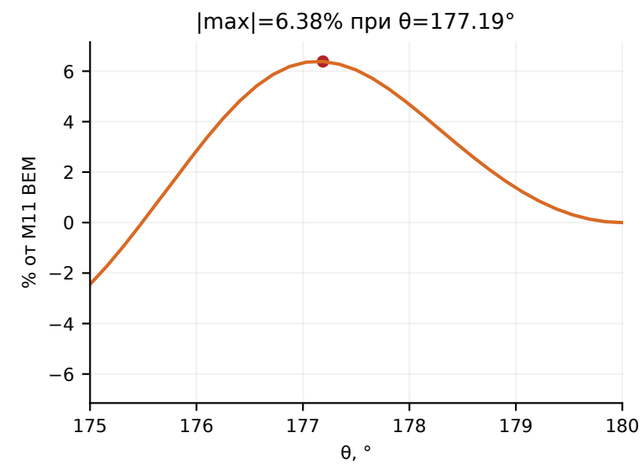


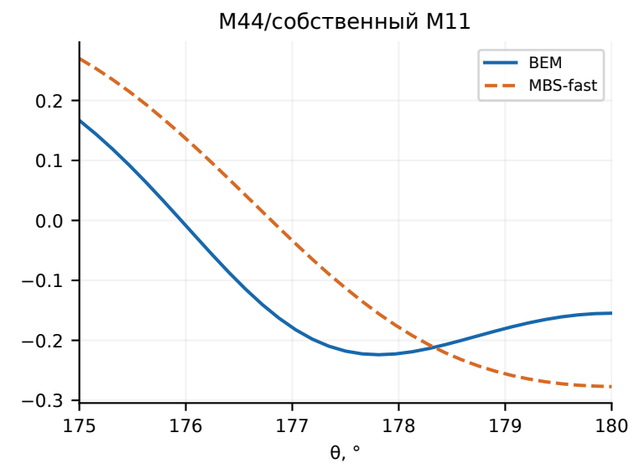


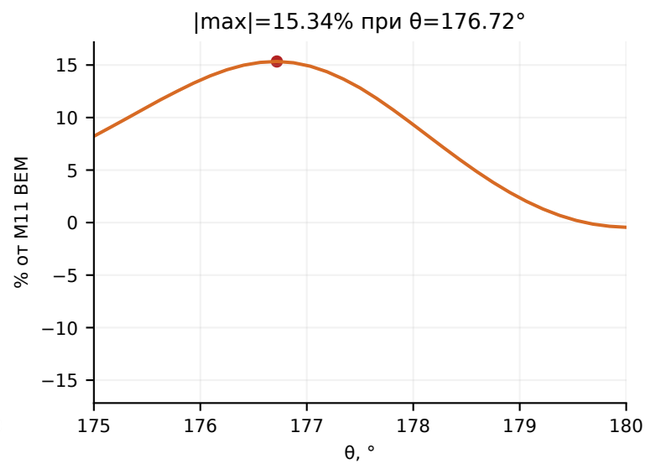












Интегральные ошибки, % — Исходные Mij

100 ∫|Q(MBS)−Q(BEM)| sinθ dθ / ∫|Q(BEM)| sinθ dθ, θ=175...180°

Элемент	ka=20	ka=25	ka=30	ka=35	ka=40
M11	23.393	22.568	13.119	17.978	15.944
M12	8.179	49.790	6.956	39.990	41.857
M13	—	—	—	—	—
M14	—	—	—	—	—
M21	18.893	62.575	23.408	63.241	51.667
M22	14.555	32.165	25.848	38.871	36.865
M23	—	—	—	—	—
M24	—	—	—	—	—
M31	—	—	—	—	—
M32	—	—	—	—	—
M33	10.983	27.154	20.417	27.683	30.454
M34	91.217	118.448	54.352	109.007	91.796
M41	—	—	—	—	—
M42	—	—	—	—	—
M43	71.953	115.657	60.883	87.644	75.794
M44	282.182	78.584	78.561	86.406	80.494

«—»: нулевой по симметрии элемент либо тождественное M11/M11.
Ошибка означает расхождение методов; допуск fullauto 1% относится к сходимости MBS.

Интегральные ошибки, % — Mij / собственный M11

100 ∫|Q(MBS)−Q(BEM)| sinθ dθ / ∫|Q(BEM)| sinθ dθ, θ=175...180°

Элемент	ka=20	ka=25	ka=30	ka=35	ka=40
M11	—	—	—	—	—
M12	31.794	40.215	10.786	30.639	36.200
M13	—	—	—	—	—
M14	—	—	—	—	—
M21	4.141	55.436	20.760	57.488	45.456
M22	12.181	15.079	16.551	25.542	24.759
M23	—	—	—	—	—
M24	—	—	—	—	—
M31	—	—	—	—	—
M32	—	—	—	—	—
M33	16.080	7.539	7.823	10.050	13.144
M34	89.182	122.379	47.769	112.047	94.247
M41	—	—	—	—	—
M42	—	—	—	—	—
M43	65.317	118.993	55.302	85.609	73.408
M44	347.783	96.960	91.347	95.992	87.682

«—»: нулевой по симметрии элемент либо тождественное M11/M11.
Ошибка означает расхождение методов; допуск fullauto 1% относится к сходимости MBS.